Control First, Robustness Next: Decoupled Representation Learning for Visual RL Generalization

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Abstract

Visual reinforcement learning (RL) is promising for real-world problems such as autonomous driving, robotic locomotion, and manipulation, but remains highly vulnerable to visual distribution shifts between training and deployment. Existing methods improve robustness through Q-consistency regularization, masking, and auxiliary objectives, yet most of them follow a coupled representation learning paradigm in which control-oriented and robustness-oriented representation learning are jointly optimized within the same training framework. We argue that this coupled structure can introduce harmful interference between control learning and robustness learning, degrading original-environment control performance and limiting generalization under visual shifts. To address this issue, we propose **Separate Then Align Representations (STAR)**, which reformulates visual RL generalization as a problem of decoupled representation learning. Specifically, we first learn a control-relevant reference representation through online RL under weak augmentation, and then perform an offline post-representation alignment stage that maps strongly augmented inputs back to this reference representation. Experiments on RL-ViGen benchmarks spanning DMControl and Robosuite show that **STAR** achieves strong generalization under visual distribution shifts while better preserving performance in the original environment than prior coupled approaches. Our source code is available in the supplementary material.

1 Introduction

Visual reinforcement learning (visual RL) has emerged as a promising approach for learning control policies directly from image observations, with applications in autonomous driving, robotic locomotion, and manipulation (**?**). However, visual RL policies often suffer from severe performance degradation when the visual distribution changes between training and evaluation, such as changes in background, lighting, object materials, or textures (**??**). Since such visual variations commonly occur in real-world environments even when the underlying task remains unchanged, robust generalization under visual distribution shift is essential for the practical deployment of visual RL.

To improve visual robustness, prior work has explored data augmentation, Q-consistency regularization, saliency- or masking-based methods, and various auxiliary objectives (**??????**). Although these methods differ in their specific designs, they commonly optimize robustness-oriented signals together with policy and value learning within the same online training process (**???**). We refer to this as a coupled representation learning paradigm, where control-oriented representation learning and robustness-oriented representation learning are co-optimized and directly shape the same learned representation. Even when weakly augmented observations are used for the RL objective and strongly augmented observations are used for consistency or auxiliary objectives, the resulting representation is still influenced by both control and robustness signals during policy optimization.

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We argue that this coupling is a central bottleneck in robust visual RL. Strong perturbations and robustness objectives introduced during policy optimization may suppress not only nuisance visual variations but also task-relevant cues required for accurate policy and value estimation. As a result, objectives intended to improve visual robustness can interfere with the formation of control-relevant representations, destabilizing RL training and degrading performance even in the original environment. This issue is particularly critical in manipulation tasks, where subtle geometric structures, object relations, and contact-relevant visual cues can significantly affect action values (**??**). Thus, the key challenge is not merely to design stronger augmentations or additional auxiliary losses, but to determine how control learning and robustness learning should be structured.

While most prior methods optimize control learning and robustness learning jointly, related work such as SECANT (**?**) has explored staged robustification after initial policy learning. However, it improves robustness through additional online policy-distillation training. In contrast, we aim to robustify a trained control policy purely at the representation level, without additional environment interaction or actor-critic retraining.

To this end, we propose **Separate Then Align Representations (STAR)**, a decoupled representation learning framework for visual RL generalization. STAR first learns a control-relevant representation through standard online RL under weak augmentation, without applying strong perturbation-based robustness objectives during policy optimization. It then freezes the learned control encoder as a fixed reference and trains a robust encoder offline to map strongly augmented observations back to this reference representation space. In this way, STAR turns robustification from a moving-target alignment problem during online RL into a fixed-target representation recovery problem after control learning. Because the robustness objective is defined over latent representations rather than policy-specific action outputs, STAR can serve as a modular post-hoc robustification method for pretrained image-based control models, including visual RL agents and imitation-based visuomotor policies such as Diffusion Policy.

We evaluate the proposed framework on DMControl and Robosuite environments in the RL-ViGen benchmark (**?**). Our results show that the proposed decoupled framework better preserves original-environment performance while improving robustness under visual distribution shifts. We further demonstrate that the same representation-level alignment principle can improve the visual robustness of diffusion policies trained from limited demonstration data.

Our contributions are summarized as follows:

- We empirically demonstrate that coupling robustness representation learning with online control learning is associated with increased representation drift and TD residuals, indicating instability in both representations and critic learning.
- We propose STAR, a decoupled visual RL framework that separates control learning from robustness learning by performing offline alignment of strongly perturbed observations to a fixed control-relevant reference representation learned through online RL.
- We demonstrate across DMControl and Robosuite that STAR improves visual generalization while better preserving original-environment performance, and further validate its extensibility to pretrained imitation-based visuomotor policies such as Diffusion Policy.

2 Background

2.1 Visual reinforcement learning

We consider a control problem defined by an underlying Markov decision process (**?**) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(s_{t+1} \mid s_t, a_t)$ is the transition probability, $r(s_t, a_t)$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. The goal is to learn a policy π that maximizes the expected discounted return $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{T} \gamma^t r(s_t, a_t)]$.

In visual reinforcement learning, the agent does not observe the underlying state s_t directly. Instead, it receives an image observation, where the same control state may appear differently due to background, lighting, texture, or camera noise. In this work, we consider the setting where policy performance degrades under visual distribution shifts even though the underlying control dynamics remain nearly unchanged. This happens because visually different observations of the same control state can

be encoded into inconsistent representations. Our goal is therefore to learn control-consistent representations that remain stable under such visual changes.

2.2 Soft actor-critic training

Our visual RL backbone follows the DrQ training scheme, which is based on Soft Actor-Critic (SAC) with weak image augmentation (**??**). Given a latent representation $z_t = E(o_t)$, the critic is trained using the soft Bellman target

$$y_t = r_t + \gamma \left(Q_{\bar{\theta}}(z_{t+1}, a_{t+1}) - \alpha \log \pi_\phi(a_{t+1} \mid z_{t+1}) \right), \qquad a_{t+1} \sim \pi_\phi(\cdot \mid z_{t+1}), \tag{1}$$

and the critic loss is

$$\mathcal{L}_{\text{critic}} = \mathbb{E}_{(o_t, a_t, r_t, o_{t+1}) \sim \mathcal{B}} \left[\left(Q_\theta(z_t, a_t) - y_t \right)^2 \right]. \tag{2}$$

The actor is trained with the entropy-regularized objective

$$\mathcal{L}_{\text{actor}} = \mathbb{E}_{o_t \sim \mathcal{B},\, a_t \sim \pi_\phi(\cdot \mid z_t)} \left[\alpha \log \pi_\phi(a_t \mid z_t) - Q_\theta(z_t, a_t) \right]. \tag{3}$$

2.3 Weak and strong visual augmentations

We use two types of visual augmentations. A *weak augmentation* f_{weak} denotes random shift, a standard augmentation in visual RL that preserves most task-relevant visual content (**?**). A *strong augmentation* f_{strong} denotes random overlay, which blends the observation with an external image (**?**) and introduces larger appearance changes in background, texture, and color (**?**). Specifically, given an observation o and an external overlay image b, random overlay constructs

$$\tilde{o} = (1 - \alpha)o + \alpha b, \qquad \alpha \sim \text{Uniform}(0, 1). \tag{4}$$

In our framework, f_{weak} is used during online control learning, while f_{strong} is used to construct perturbed observations for post-representation alignment.

3 Motivation

Many visual RL generalization methods jointly optimize control learning and robustness-oriented representation learning. However, the agent must learn both what is relevant for control and what should remain invariant to visual perturbations from scratch. This joint learning can be challenging because the two objectives are optimized while influencing the same representation. In actor-critic RL, the control representation is formed together with the critic through bootstrapped updates, while robustness objectives simultaneously encourage invariance through consistency, regularization, or auxiliary losses (**?????**).

Following standard visual RL practice, the visual encoder is updated through the critic loss and kept frozen during actor updates. Therefore, the learned representation is shaped primarily by the critic-side objective (**??**).

Representation and TD-learning changes under coupled learning Figure 1 compares source-domain performance and learning dynamics under coupled robustness learning. We measure *representation shift* as the change in the latent embedding of the same observations over training, and *normalized TD error* as the mismatch between the current Q-value and its Bellman target relative to the target magnitude. DrQ (**?**) achieves higher average source-domain return than SVEA (**?**), SMG (**?**), and SimGRL (**?**) on both DMC and Robosuite.

The coupled methods also exhibit larger representation shifts and normalized TD errors than DrQ. On DMC, the two quantities reach 1.24–$2.05\times$ and 1.34–$1.65\times$ the DrQ levels, respectively; on Robosuite, they increase up to $4.54\times$ and $5.57\times$. Thus, the differences under coupled learning are reflected not only in the learned representation but also in TD learning.

A key difficulty is that robustness is optimized while the control representation is still evolving through online RL. Representations induced by strongly augmented observations must therefore adapt both to visual perturbations and to a changing control representation. This motivates a simple principle: *control first, robustness next*. STAR first establishes the representation needed for control and then learns robustness relative to this fixed representation.

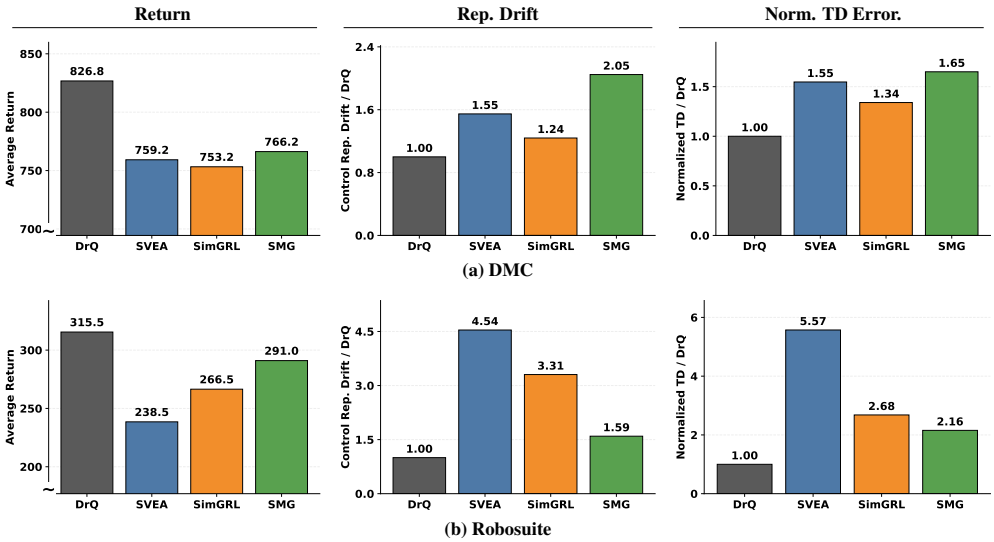


Figure 1: From left to right: in-distribution return, representation shift, and normalized TD Error. Representation shift and normalized TD Error are reported relative to the corresponding DrQ value ($= 1.0$).

4 Method

STAR consists of two stages. Let E_C denote the encoder that produces the latent representation used for control. In Stage 1, E_C is trained through DrQ-style online RL with an auxiliary forward-dynamics objective to learn a control-relevant reference representation.

After Stage 1, the reference encoder and downstream controller are frozen. In Stage 2, a robust encoder E_R, initialized from E_C, and an alignment module g_ϕ are trained offline to recover the fixed reference representation from strongly augmented observations. Only the robust visual pathway is updated during this stage.

We refer to this Stage-2 procedure as *Post-Representation Alignment* (PRA), while STAR denotes the complete two-stage framework.

4.1 Stage 1: control-oriented reference learning

In Stage 1, we train a visual RL agent in the source environment following the DrQ-style (**?**) training pipeline. We do not apply strong perturbation-based robustness objectives during this stage. The purpose of Stage 1 is to learn a stable control representation that can later serve as the reference for post-representation alignment.

Let E_C denote the encoder, including the visual encoder and projection module. Given an observation o_t, the encoder produces a latent representation

$$z_t^C = E_C(o_t),$$

which is used by both the actor and critic for policy and value learning.

In addition to the standard actor-critic losses, we add a lightweight forward dynamics objective to encourage the latent representation to capture action-conditioned transition information given a transition (o_t, a_t, o_{t+1}), the dynamics model predicts the next latent representation:

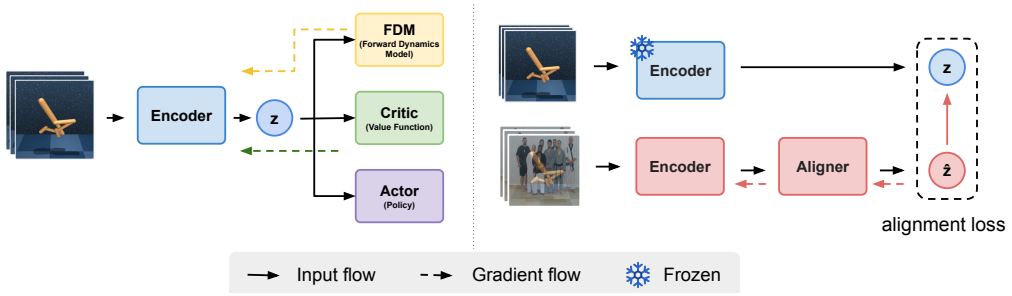
$$\mathcal{L}_{\text{dyn}} = \mathbb{E}_{\mathcal{B}} \left[\left\| F_\psi(z_t^C, a_t) - \text{sg}[z_{t+1}^C] \right\|_2^2 \right], \tag{5}$$

where $z_{t+1}^C = E_C(o_{t+1})$ and $\text{sg}[\cdot]$ denotes stop-gradient.

The Stage 1 representation objective is

$$\mathcal{L}_{\text{ctrl-rep}} = \mathcal{L}_{\text{critic}} + \lambda \mathcal{L}_{\text{dyn}}. \tag{6}$$

After Stage 1, E_C is frozen, and its output z_t^C is used as the target reference latent.



(a) Stage 1: online control-oriented learning. (b) Stage 2: offline post-representation alignment.

Figure 2: Overview of the proposed two-stage framework. Stage 1 learns the reference latent z using forward dynamics and critic objectives to capture transition- and value-relevant information. Stage 2 freezes the pretrained actor and critic and aligns the shifted latent $\hat{z}$ with z.

4.2 Stage 2: offline post-representation alignment

After online RL-based control learning, we perform robustification without any additional environment interaction. We use only observations from the replay buffer $\mathcal{D}$ collected during Stage 1. The control encoder E_C is frozen and serves as the reference encoder, while a robust encoder E_R is initialized from E_C and updated during alignment.

For each observation $o_t \sim \mathcal{D}$, we construct a weakly augmented view $o_t^{\mathrm{w}} = f_{\mathrm{weak}}(o_t)$ and a strongly perturbed view $\tilde{o}_t = f_{\mathrm{strong}}(o_t)$. The weak augmentation uses the same random shift as in Stage 1, while the strong augmentation introduces diverse visual perturbations. We obtain a reference representation from the weak view using the frozen control encoder E_C, and an aligned robust representation from the strongly perturbed view using E_R and g_ϕ:

$$z_t^{\mathrm{ref}} = \mathrm{sg}[E_C(o_t^{\mathrm{w}})]\,, \quad z_t^R = g_\phi(E_R(\tilde{o}_t))\,, \tag{7}$$

where $\mathrm{sg}[\cdot]$ denotes stop-gradient and g_ϕ is a lightweight alignment module. We train E_R and g_ϕ to align the robust representation with the fixed control-relevant reference by minimizing

$$\mathcal{L}_{\mathrm{align}} = \mathbb{E}_{o_t \sim \mathcal{D}} \left[\left\| z_t^R - z_t^{\mathrm{ref}} \right\|_2^2 \right]. \tag{8}$$

For the strong augmentation f_{strong}, we use the random overlay defined in Eq. (4). Since the mixing coefficient α is sampled for each observation; the robust encoder is exposed to a range of perturbation strengths, from mild appearance changes to severe visual corruptions. This avoids relying on a single fixed overlay ratio and encourages the robust encoder to align perturbed observations of varying difficulty to the same reference latent.

During Stage 2, only the robust encoder E_R and the alignment module g_ϕ are updated. The reference encoder E_C, actor, critic, and optional forward dynamics module remain fixed. Thus, robustification is performed purely at the representation level: strongly perturbed observations are mapped back to the fixed control-relevant latent learned during Stage 1, without policy or critic retraining.

4.3 Inference with the Fixed Controller

As illustrated in Figure 2(b), Stage 2 trains the robust encoder E_R and alignment module g_ϕ to recover the latent representation produced by the Stage-1 reference encoder E_C from strongly augmented observations. At evaluation time, the robust pathway $g_\phi \circ E_R$ replaces E_C, while the learned controller remains fixed.

At inference, an observation o_t is mapped by the robust pathway to the aligned representation $z_t^R = g_\phi(E_R(o_t))$, which is passed directly to the frozen controller. The reference encoder E_C is used only to construct alignment targets during Stage 2 and is not required at inference.

5 Experiments

5.1 Evaluation questions

We evaluate the proposed *decoupled representation learning* framework from three complementary perspectives. Specifically, our experiments are designed to answer the following questions.

Q1. Effectiveness of decoupled representation learning. Can the proposed decoupled representation learning framework improve visual robustness under distribution shifts while preserving performance in the original environment?

Q2. Performance on contact-rich manipulation. Does the proposed method achieve stronger performance than prior robust visual RL baselines on contact-rich manipulation tasks?

Q3. Roles of Dynamics Supervision and Stage-2 Alignment. What are the respective contributions of Stage-1 dynamics supervision and Stage-2 post-representation alignment?

Q4. Policy-class extensibility. Can the proposed method be applied not only to pretrained visual RL policies but also to other policy classes, including imitation-based policies such as Diffusion Policy?

Together, these questions allow us to assess whether the proposed decoupled representation learning framework can effectively improve visual robustness, remain competitive on contact-rich manipulation tasks, and extend to diverse policy classes.

5.2 Experimental setup

5.2.1 Baselines

We compare **STAR** with representative robust visual RL baselines. **DrQ** (**?**) is the vanilla baseline, using random shift without an explicit robustness objective. **SVEA** (**?**), **SGQN** (**?**), **SMG** (**?**), and **SimGRL** (**?**) introduce robustness-oriented objectives during online RL, such as value consistency, saliency guidance, reconstruction or feature consistency, and strong visual augmentation. We also compare with **SECANT** (**?**), a staged robustification baseline that first trains a teacher policy and then learns a robust student policy in a second stage.

5.2.2 Benchmarks and metrics

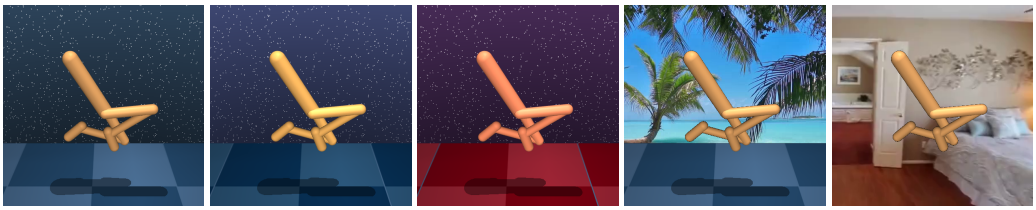
We evaluate **STAR** on the RL-ViGen benchmark (**?**), which tests visual generalization under distribution shifts such as changes in background, color, texture, lighting, and object appearance. We consider two environment groups: DMC-GB, based on the DeepMind Control Suite (**?**), and Robosuite manipulation tasks (**?**).

For online RL experiments, we evaluate on DMC-GB tasks including *walker walk*, *cheetah run*, *finger spin*, and *cartpole swingup*, as well as Robosuite tasks including *Door* and *TwoArmPegInHole*. These policies are trained in the original environment and evaluated zero-shot under visual distribution shifts. We report episodic return as the evaluation metric.

To evaluate the extensibility of representation-level alignment beyond online RL policies, we also consider pretrained imitation-based visuomotor policies. Specifically, we train Diffusion Policy models (**?**) from expert demonstrations on two Robosuite tasks, *PickPlaceCan* and *NutAssemblySquare*, and apply our offline alignment objective to their visual representations. For these diffusion policy experiments, we report the task success rate under visual distribution shifts.

5.3 Q1: Effectiveness of decoupled representation learning

Overall, **STAR** not only achieves the best performance across all four DMC-GB tasks in the Original setting, but also shows strong zero-shot generalization under visual distribution shifts. Across the shifted settings, it achieves the best or competitive performance in most cases. The gains are particularly clear in tasks where **STAR** learns a stronger source-domain policy. For example, in *Walker Walk* and *Cheetah Run*, **STAR** shows relatively large improvements in the Original setting, and this advantage often carries over to shifted environments such as Color Easy, Color Hard, and



(a) Training (b) Color-easy (c) Color-hard (d) Video-easy (e) Video-hard

Figure 3: Examples of training and evaluation observations for Walker Walk in RL-ViGen DMC-GB. Color Easy and Color Hard change the scene appearance. Video Easy and Video Hard introduce video backgrounds; Video Hard also removes the floor and the walker's shadow.

Video Easy. This suggests that learning a stronger control-relevant reference representation in Stage 1 can provide a better target for subsequent representation alignment.

While **STAR** is not uniformly best under every shift, it remains highly competitive. In a few cases, such as *Finger Spin* and *Cartpole SwingUp* under Video Easy or *Cheetah Run* under Video Hard, it is slightly behind the best baseline. Overall, however, **STAR** preserves strong Original performance while achieving robust zero-shot generalization across diverse DMC-GB shifts.

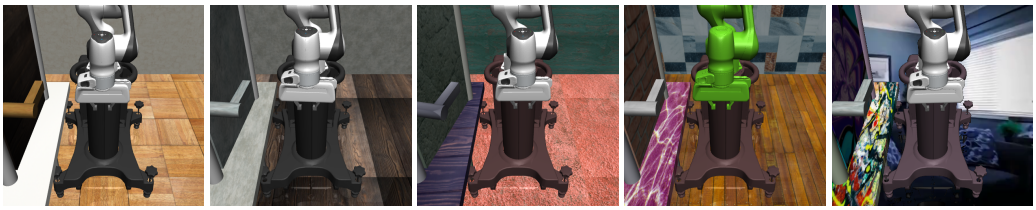
Table 1: Performance comparison on RL-ViGen DMC-GB. We report mean episodic return ± standard deviation over four seeds. The best result in each task–setting pair is shown in bold.

Setting	Task	DrQ	SVEA	SGQN	SMG	SimGRL	SECANT	**STAR**	Δ
Original	Walker	932±19	851±78	895±34	877±93	870±78	904±21	**958±6**	+26 (3%)
	Cheetah	524±127	354±32	284±21	361±22	310±38	460±84	**615±52**	+91 (17%)
	Finger	983±2	968±24	984±1	979±3	972±22	956±21	**985±1**	+1 (1%)
	Cartpole	868±12	864±12	861±7	848±7	861±12	860±12	**872±11**	+4 (1%)
Color Easy	Walker	772±145	821±44	856±62	873±44	799±101	875±45	**939±15**	+64 (7%)
	Cheetah	378±47	313±46	253±30	358±11	302±41	423±75	**567±64**	+144 (34%)
	Finger	950±34	888±178	921±29	971±5	972±8	946±27	**975±10**	+3 (1%)
	Cartpole	770±58	828±48	788±43	843±11	860±16	858±13	**869±15**	+9 (1%)
Color Hard	Walker	615±87	781±35	718±53	806±49	682±92	800±48	**880±43**	+74 (9%)
	Cheetah	251±38	299±39	229±18	337±13	305±31	388±70	**493±91**	+105 (27%)
	Finger	813±109	879±183	853±46	912±15	907±13	**939±15**	907±41	−32 (3%)
	Cartpole	551±129	802±51	743±33	811±9	812±27	809±17	**824±34**	+12 (1%)
Video Easy	Walker	709±76	802±80	860±35	863±89	847±81	803±38	**950±8**	+87 (10%)
	Cheetah	268±62	277±25	242±3	333±33	302±42	264±76	**510±87**	+177 (53%)
	Finger	538±59	724±20	946±22	966±17	947±24	672±43	**980±7**	+14 (1%)
	Cartpole	424±49	811±17	769±60	787±18	**839±16**	721±49	837±40	−2 (1%)
Video Hard	Walker	119±56	417±96	732±56	802±103	795±97	331±34	**924±18**	+122 (15%)
	Cheetah	60±6	104±36	189±21	**310±17**	303±46	77±26	271±67	−39 (13%)
	Finger	7±4	242±40	781±93	925±39	927±17	174±28	**977±8**	+50 (5%)
	Cartpole	133±30	325±57	644±46	405±115	746±69	317±39	**828±21**	+82 (11%)
Original avg.		826.8	759.3	756.0	766.3	753.3	795.0	**857.5**	+30.7 (4%)
Shifted avg.		459.9	582.1	657.8	706.4	709.1	587.3	**795.7**	+86.6 (12%)

5.4 Q2: Performance on contact-rich manipulation

We next evaluate whether **STAR** remains effective in contact-rich manipulation tasks, where robust perception must be combined with precise object interaction. We use two Robosuite tasks from RL-ViGen, *Door* and *TwoArmPegInHole*, under a 500K online interaction budget. Compared with DMC-GB, these tasks involve more structured scene geometry and contact-sensitive control, making them a challenging testbed for visual robustness.

Table 2 reports the results. The proposed method achieves the best performance on both tasks in the Original setting, indicating that decoupled representation learning preserves strong source-domain



(a) Training (b) Easy (c) Medium (d) Hard (e) Video

Figure 4: Examples of training and evaluation observations for RL-ViGen Robosuite tasks. The panels use the Original, Easy, Medium, Hard, and Video labels adopted in this paper; their mapping to the benchmark settings is described in the text.

control performance even in manipulation environments. Under visual distribution shifts, it also obtains the best performance in 7 out of 8 shifted settings. The gains are especially large on *Door*, where our method consistently outperforms all baselines across Easy, Medium, Hard, and Video shifts.

On *TwoArmPegInHole*, the proposed method also shows strong robustness, achieving the best performance in Easy, Medium, and Video settings. The only exception is the Hard setting, where SECANT performs slightly better. Nevertheless, our method remains competitive in this case and substantially outperforms the other coupled robustness baselines.

Overall, these results show that the proposed framework is not limited to locomotion or simple continuous-control tasks. By preserving a strong control-relevant representation and aligning perturbed observations to it, our method provides robust zero-shot generalization in contact-rich manipulation environments.

Table 2: Performance comparison on RL-ViGen Robosuite manipulation tasks. We report mean episodic return ± standard deviation over four seeds. The best result in each task–setting pair is shown in bold. TwoArm denotes TwoArmPegInHole.

Setting	Task	DrQ	SVEA	SGQN	SMG	SimGRL	SECANT	**STAR**	Δ
Original	TwoArm	399±84	303±189	282±43	300±92	334±127	415±95	**473±6**	+58 (14%)
	Door	232±221	174±188	279±151	282±172	199±250	253±200	**455±32**	+173 (61%)
Easy	TwoArm	259±45	282±165	281±40	297±82	317±110	397±76	**461±13**	+64 (16%)
	Door	48±33	144±153	180±86	237±172	165±209	250±184	**440±34**	+190 (76%)
Medium	TwoArm	163±26	259±133	273±27	287±83	296±77	370±76	**397±22**	+27 (7%)
	Door	12±8	118±118	146±80	129±89	86±103	225±171	**442±38**	+217 (96%)
Hard	TwoArm	149±10	214±92	251±14	257±68	201±42	**357±79**	335±57	−22 (6%)
	Door	10±16	62±81	119±93	138±142	46±69	167±166	**202±145**	+35 (21%)
Video	TwoArm	143±24	152±20	205±4	168±13	265±87	243±25	**424±9**	+159 (60%)
	Door	2±2	40±39	100±53	51±23	24±19	47±28	**343±41**	+243 (243%)
Original avg.		315.5	238.5	280.5	291.0	266.5	334.0	**464.0**	+130.0 (39%)
Shifted avg.		98.3	158.9	194.4	195.5	175.0	257.0	**380.5**	+123.5 (48%)

5.5 Q3: Roles of Dynamics Supervision and Stage-2 Alignment

We conduct an ablation study to examine the roles of Stage-1 dynamics supervision (Dyn) and Stage-2 post-representation alignment (S2). Table 3 reveals a clear separation between their contributions. Stage 2 provides the primary gain in visual robustness. Without dynamics supervision, adding S2 increases the shifted average from 399.9 to 773.4 on DMC and from 96.0 to 262.0 on Robosuite, while leaving Original performance nearly unchanged. This shows that post-representation alignment remains effective even when the Stage-1 representation is learned without dynamics supervision.

Dynamics supervision plays a complementary role. Without S2, adding Dyn has little effect on shifted performance (399.9 vs. 399.3 on DMC and 96.0 vs. 103.8 on Robosuite), but substantially improves

Table 3: Matched ablation of Stage-1 dynamics supervision (Dyn) and Stage-2 post-representation alignment (S2). Values are average episodic returns; Shifted averages over all appearance-shift conditions in each benchmark.

		w/o S2		w/ S2	
Benchmark	Evaluation	−Dyn	+Dyn	−Dyn	+Dyn
DMC	Original	852.5	**858.0**	850.3	857.5
	Shifted	399.9	399.3	773.4	**795.7**
Robosuite	Original	319.5	**464.5**	315.5	464.0
	Shifted	96.0	103.8	262.0	**380.5**

Original performance on Robosuite from 319.5 to 464.5. When combined with S2, Dyn further increases the shifted average from 773.4 to 795.7 on DMC and from 262.0 to 380.5 on Robosuite.

These results separate the two roles in STAR: Stage 1 establishes a control-relevant reference, while Stage 2 provides visual robustness by recovering this fixed reference from strongly augmented observations. Dynamics supervision improves the learned control reference, particularly on Robosuite, whereas Stage 2 is the direct mechanism for post-hoc robustification. Full task-level results are provided in Appendix **??**.

5.6 Q4: Policy-class extensibility

The third question asks whether the proposed framework can extend beyond actor–critic-based visual RL policies to other policy classes. To this end, we apply post-representation alignment to a pretrained Diffusion Policy (**?**). This experiment tests whether robustness can be improved by aligning visual representations alone, without retraining the diffusion-based action generation module.

We evaluate this setting under visual distribution shifts constructed by modifying scene appearance, such as textures, object appearance, and background visuals, while keeping the underlying task unchanged. Details of these visual modifications are provided in Appendix **??**.

Concretely, we initialize the robust visual encoder from the pretrained Diffusion Policy encoder and apply the same offline representation alignment objective.

After alignment, the robust encoder module is connected to the pretrained diffusion action module for evaluation. The results in Table 4 show that post-representation alignment improves the robustness of Diffusion Policy under visual distribution shifts.

These findings support our broader claim that robustness learning can be separated from policy learning. Since the action generation module is kept fixed, the improvement comes from aligning perturbed observations to the latent interface expected by the pretrained policy, suggesting that the proposed framework can serve as a general post-representation robustification mechanism for visuomotor policies.

Table 4: Success rates (%) of Diffusion Policy with and without post-representation alignment (PRA). Easy, Medium, and Hard denote increasing visual-shift severity.

Task	Method	Original	Easy	Medium	Hard
Can	DP	$\mathbf{100.0 \pm 0.0}$	39.3 ± 4.1	22.0 ± 4.3	14.0 ± 3.3
	DP + PRA	99.3 ± 0.9	$\mathbf{72.0 \pm 3.3}$	$\mathbf{62.0 \pm 4.3}$	$\mathbf{41.3 \pm 7.4}$
Square	DP	96.0 ± 1.6	43.3 ± 3.4	28.0 ± 1.6	15.3 ± 2.5
	DP + PRA	$\mathbf{98.0 \pm 0.0}$	$\mathbf{76.0 \pm 2.8}$	$\mathbf{74.7 \pm 5.2}$	$\mathbf{67.3 \pm 0.9}$

6 Related work

Representation learning for visual RL. Visual RL must learn control from high-dimensional image observations while improving both sample efficiency and visual generalization. Prior work

improves visual representations through auxiliary objectives, including contrastive learning methods (**??**), dynamics-based prediction methods (**???**), and broader self-supervised objectives such as reconstruction, future prediction, or latent world modeling (**??**). Another line focuses on robustness to visual shifts in background, color, texture, lighting, and object appearance through augmentation, consistency regularization, masking, saliency guidance, representation separation, or auxiliary losses (**??????**). Despite their differences, these methods generally couple representation learning, robustness, and policy/value optimization within the same training loop.

SECANT (**?**) is the closest prior work because it also separates policy learning and robustification into two stages. However, SECANT trains a student policy through DAgger-style (**?**) online imitation under strong augmentation, whereas **STAR** freezes the Stage 1 control representation and performs purely offline post-representation alignment, without rollouts and actor–critic retraining.

7 Discussion

Our ablations show that STAR's two stages serve complementary roles: Stage-1 establishes a strong control-relevant representation, while Stage-2 recovers it under visual perturbations to provide the main robustness gain. This separation enables lightweight post-hoc robustification without additional environment interaction or downstream controller retraining, and applies to independently pretrained DrQ and Diffusion Policy. These results suggest a path toward broader pretrained visuomotor policies, including Vision-Language-Action models.

Limitations. Despite the promising results, our framework has several limitations. First, the effectiveness of Stage 2 depends on the quality of the control-relevant representation learned in Stage 1. If the Stage 1 policy learns a suboptimal representation, the alignment stage may inherit this limitation rather than correct it. Second, when applied to imitation-based policies such as Diffusion Policy, post-representation alignment improves shifted-environment performance but can slightly degrade original-environment performance, suggesting a remaining trade-off between robustness and preserving the original policy behavior. Finally, following prior visual generalization studies, our scope assumes that visual shifts mainly affect the observation appearance, while the underlying control problem remains unchanged. If the environment dynamics or task semantics change, additional techniques such as domain randomization or domain adaptation may be required.

Future Work. Future work can improve both the formation and recovery of control-relevant representations. Stronger control-oriented objectives may provide better Stage-1 references, while control-aware alignment could selectively recover features or subspaces that matter most for downstream control rather than matching the entire latent uniformly. Extending this principle to broader visual shifts and larger pretrained visuomotor and Vision-Language-Action models is a promising direction.

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